Laws of Indices

Number · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 110 minutes · Total marks: 110

KEY RULES

Multiply, same base	$a^m \times a^n = a^{m+n}$
Divide, same base	$a^m \div a^n = a^{m-n}$
Power of a power	$(a^m)^n = a^{mn}$
Zero and negative powers	$a^0 = 1\ ;\ a^{-n} = \frac{1}{a^n}$
Fractional powers	$a^{\frac{1}{n}} = \sqrt[n]{a}\ ;\ a^{\frac{m}{n}} = (\sqrt[n]{a})^m$

WORKED EXAMPLE 1

[2 marks]

Simplify $a^5 \times a^3$

Step 1	Same base: add the powers	a^{5+3}

Answer: a^8

WORKED EXAMPLE 2

[2 marks]

Simplify $(2x^3)^4$

Step 1	Raise the number to the power 4	$2^4 = 16$
Step 2	Multiply the powers	$(x^3)^4 = x^{12}$

Answer: $16x^{12}$

WORKED EXAMPLE 3

[3 marks]

Work out $8^{\frac{2}{3}}$

Step 1	The denominator 3 means cube root	$\sqrt[3]{8} = 2$
Step 2	The numerator 2 means square	$2^2 = 4$

Answer: 4

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1 Write 5^{-2} as a fraction.

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

2 Work out $64^{\frac{1}{3}}$

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

3 Work out $27^{\frac{2}{3}}$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

4 Work out $\left(\frac{1}{4}\right)^{-2}$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

5 Work out $16^{-\frac{1}{2}}$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

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6 Simplify $(27x^6)^{\frac{1}{3}}$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

7 Simplify $2x^2y \times 3xy^4$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

8 Write $\frac{1}{\sqrt{x}}$ as a power of x.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

9 Solve $n^4 = 16$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

10 Simplify $\left(4x^5y^6\right)^{-\frac{1}{2}}$

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11 7 $\frac{3^{-2} \times 3^5}{3^{10}} = 3^n$

Find the value of n

$n =$ .....

(Total for Question 7 is 2 marks)

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12 7 (a) Simplify $(4^{-2})^0$

..... (1)

$3^{-14} \times 3^8 = 3^m$

(b) Find the value of m

$m =$ ..... (1)

(Total for Question 7 is 2 marks)

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13 8 $\frac{8^{-2} \times 8^9}{8^{10}} = 8^n$

Work out the value of n

$n =$ .....

(Total for Question 8 is 2 marks)

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14 Given that $150^x = 1$

(a) write down the value of x.

$x =$ .....
(1)

Given that $3^{-8} \div 3^{-6} = 3^n$

(b) find the value of n.

$n =$ .....
(1)

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15 **12** $3^{\frac{1}{2}} \times 3^{\frac{2}{5}} = 3^m$

(a) Work out the value of m

$m =$ .....
(1)

$^{-10} \div 5^{-4} = 5^n$

b) Work out the value of n

$n =$ .....
(1)

(Total for Question 12 is 2 marks)

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16 Solve $3^x = \frac{1}{9}$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

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17 Solve $2^{x+1} = 32$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

18 Solve $3 \times 4^x = 48$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

19 Given that $\frac{3^x}{9^{3x}} = 81$

find the value of x.
Show clear algebraic working.

$x =$ .....

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20 **14** Given that $\frac{3^{2n+3}}{3^4} = 3^3 \times 3^{1-2n}$

find the value of n
Show your working clearly.

$n =$ .....

(Total for Question 14 is 3 marks)

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21 (a) Write down the value of m, given that $3^4 \times 3^5 = 3^m$

$m =$ .....
(1)

(b) Write down the value of n, given that $(5^3)^7 = 5^n$

$n =$ .....
(1)

(c) Find the value of p, given that $\frac{7^8 \times 7^2}{7^p} = 7^6$

$p =$ .....
(2)

(Total for Question 1 is 4 marks)

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22 $x^5 \times x^7 = x^m$

(a) Find the value of m

$m =$ .....

(1)

$y^3 = y^n$

Find the value of n

$n =$ .....

(1)

Simplify fully $(5a^4r^2)^3$

.....

(2)

(Total for Question 4 is 4 marks)

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23 (a) Simplify $(3k^2)^4$

.....

(2)

(b) Simplify $(21m^4n) \div (3n^{-5})$

.....

(2)

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24 (a) Simplify $m^{10} \div m^3$

.....

(1)

$\ldots^n \times k^4 = k^{12}$

b) Write down the value of n

$n =$ .....

(1)

Simplify $(3x^6y^8)^2$

.....

(2)

(Total for Question 6 is 4 marks)

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25 **6** (a) Write down the value of x^0

.....

(1)

Given that $2^{-3} \times 2^9 = 2^n$

(b) find the value of n

$n =$ .....

(1)

Given that $\frac{7^{206} \times 7^m}{7^{214}} = 7^{-3}$

(c) find the value of m

$m =$ .....

(2)

(Total for Question 6 is 4 marks)

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26 (a) Simplify $8 \times (4t)^0$

.....

(1)

$x^6 \div x^{-5} = x^p$

(b) Find the value of p

$p =$ .....

(1)

Simplify fully $(2k^2m^4)^3$

.....

(2)

(Total for Question 9 is 4 marks)

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27 (a) Simplify $(2p)^0$ where $p > 0$

.....

(1)

$y^9 \times y^{-3} = y^n$

b) Find the value of n

$n =$ .....

(1)

Simplify fully $(5a^4c^2)^3$

.....

(2)

(Total for Question 10 is 4 marks)

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28 **11** $\sqrt{2} \times 16 = 2^x$

(a) Find the value of x.
Show your working clearly.

$x =$ .....

(2)

$\frac{11^{-6})^5}{11^4} = 11^n$

b) Find the value of n.
Show your working clearly.

$n =$ .....

(2)

(Total for Question 11 is 4 marks)

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29 $2^7 \times 4^5 = 4^x$

(a) Calculate the value of x

$x =$ .....

(2)

Simplify fully $\left(125p^6y^{24}\right)^{\frac{2}{3}}$

.....

(2)

(Total for Question 12 is 4 marks)

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30 **13** (a) Simplify fully $\left(\frac{125x^6}{y^{15}}\right)^{\frac{1}{3}}$

.....

(2)

Given that $4^n = \frac{4^m}{64^p}$

(b) Express n in terms of m and p

$n =$ .....

(2)

(Total for Question 13 is 4 marks)

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31 (a) Write down the value of 5^0

.....

(1)

$$\frac{5^9 \times 5^{-3}}{5^{-2}} = 5^k$$

b) Find the value of k

$k =$ .....

(2)

Simplify fully $\left(2d^4e^5\right)^3$

.....

(2)

(Total for Question 6 is 5 marks)

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CHALLENGE QUESTIONS

32 Given that

$$2^n = 2^{x^2} \times 16^x \times 8$$

and

$$x > 0$$

find an expression for x in terms of n

State any restrictions on n

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33 Express $\left(\frac{m^6k^{10}}{25}\right)^{\frac{3}{2}}$ in the form $\frac{m^ak^b}{c}$ where a, b and c are integers to be found

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34 Simplify $\dfrac{30 \times 25^{2x+7}}{\sqrt{180} \times \left(\sqrt{5}\right)^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

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35 Given that $M = \dfrac{18^{4n} \times 2^{3(n^2-6n)} \times 3^{2(1-4n)}}{12^2}$

find the values of n for which $M = 2$

.....

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36 (a) Simplify $8^2 \times \sqrt[3]{4^6}$

Give your answer in the form 2^a where a is an integer.

Show each stage of your working clearly.

.....

(3)

Given that $n^{\left(-\frac{4}{5}\right)} = \left(\frac{1}{2}\right)^4$ where $n > 0$

(b) find the value of n.

$n =$ .....

(4)

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TOTAL FOR PAPER IS 110 MARKS